| Program : **Diploma in Chemical Engineering** | |
|---|---|
| Course Code : **5073C** | Course Title: **Material Science and Technology** |
| Semester : **5** | Credits: **4** |
| Course Category: **Program Core** | |
| Periods per week: **4 (L: 4 T: 0 P: 0)** | Periods per semester: **60** |

**Course Objectives:**

- To impart the basic concepts of metallurgy and materials of construction in chemical engineering

**Course Pre-requisites:**

| Topic | Course code | Course name | Semester |
|---|---|---|---|
| Atom and atomic structure | | Applied Chemistry I | 1 |

**Course Outcomes :**

On completion of the course student will be able to:

| COn | Description | Duration (Hours) | Cognitive Level |
|---|---|---|---|
| CO1 | Summarize the structural properties of crystals | 13 | Understanding |
| CO2 | Summarize the properties of materials | 17 | Understanding |
| CO3 | Summarize the processing and application of metal alloys | 17 | Understanding |
| CO4 | Summarize the characteristics and processing of inorganic materials | 13 | Understanding |

**CO – PO Mapping**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|---|---|---|---|---|---|---|---|
| **CO1** | 2 | | | | | | |
| **CO2** | 2 | | | | | | |
| **CO3** | 2 | | | | | | |
| **CO4** | 2 | | | | | | |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline**

| Module Outcome | Description | Duration (Hours) | Cognitive Level |
|---|---|---|---|
| **CO1** | **Summarize the structural properties of crystals** | | |
| M1.01 | Outline the importance of material science and technology. | 2 | Understanding |
| M1.02 | Identify the various materials and its applications | 3 | Understanding. |
| M1.03 | Outline crystal structure | 4 | Understanding. |
| M1.04 | Summarize the imperfections in crystal. | 4 | Understanding |
| **Contents:**<br>Material science engineering: necessity for material science and engineering- components of material science and engineering.<br>Classification of materials with applications- engineering requirement of materials -selection of materials.<br>Crystal structure: structure of atoms and molecules -Crystal structure- Crystal geometry, structure of solids, methods of determining structures-X-ray diffraction technique-nano crystalline solids.<br>Imperfection in crystals: types of imperfection. Point imperfection and types-miscellaneous imperfections and types - microscopic examination; optical microscopy, SEM, TEM | | | |
| **CO2** | **Summarize the properties of materials** | | |
| M2.01 | Summarise the mechanical properties of materials | 5 | Understanding |
| M2.02 | Summarise the elastic and hardness properties of material | 4 | Understanding |
| M2.03 | Summarise the electrical properties of the material. | 4 | Understanding |
| M2.04 | Summarise the magnetic properties of the material | 3 | Understanding |
| | Series Test – I | 1 | |

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|---|---|---|---|
| **Contents:**<br>Properties of materials: Mechanical; stress and strain, tension, compression, shear and torsional test-stress strain behavior.<br>Elastic properties: plastic deformation- tensile properties, yielding and yield strength, tensile strength, ductility, resilience, toughness- hardness; Rockwell hardness, Brinell hardness, Vickers hardness.<br>Electrical properties: electrical conduction and conductivity-superconductivity-semi conductivity- electrical conduction in polymers and ionic ceramics- dielectric behavior – ferro electricty – Piezo electricity.<br>Magnetic properties: magnetism- types of magnetism- Dia-magnetism- Para-magnetism- Ferro-magnetism- Anti-ferro-magnetism- Ferri-magnetism-with examples. | | | |
| **CO3** | **Summarize the processing and application of metal alloys** | | |
| M3.01 | Summarise the processing of metal alloys | 4 | Understanding |
| M3.02 | Summarise the thermal processing of metal alloys | 4 | Understanding |
| M3.03 | Interpret corrosion in metal and alloys. | 4 | Understanding |
| M3.04 | Summarize the properties and application of metal and alloys | 5 | Understanding |
| **Contents:**<br>Processing of metal alloys: types of alloys-ferrous and nonferrous alloys-forming operation and types- casting operation and types- powder metallurgy-welding.<br>Thermal processing- annealing-heat treatment of steel-precipitation hardening.<br>Corrosion of metals: electro chemical consideration-corrosion rates – prediction of corrosion-forms of corrosion – corrosion prevention methods.<br>Properties and application of Metals and alloys: Engineering materials - ferrous metals - Iron and their alloys Iron and steel - Iron carbon equilibrium diagram. Non-ferrous metals and alloys - Aluminum, copper, Zinc, lead, Nickel and their alloys with reference to the application in chemical industries. | | | |
| **CO4** | **Summarize the characteristics and processing of inorganic materials** | | |
| M4.01 | Outline the characteristics and processing of ceramics | 3 | Understanding |
| M4.02 | Outline the characteristics and processing of polymers. | 3 | Understanding |
| M4.03 | Outline the characteristics and processing of composites. | 3 | Understanding |
| M4.04 | Identify the types and applications of inorganic materials. | 3 | Understanding |
| | Series Test – II | 1 | |

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| **Contents:**<br>Characteristic and processing of ceramics: ceramics – crystal structure - silicates and carbon- mechanical properties – ceramic fabrication techniques - glass forming - particulate forming- cementation.<br>Characteristic and processing of polymers: polymer classification- Molecular characteristics (classification only) - polymerization- polymer additives- forming techniques for plastics, elastomers and fibers - Mechanical behavior- crystallization, melting and glass transition in polymers- degradation of polymers.<br>Characteristic and processing of composites: composite- composite types- processing of fiber reinforced composites- structural composites.<br>Types and application of inorganic materials: Ceramics- types; glass, clay products, refractories, abrasives, advanced ceramics. Ceramics, Glass and refractories – polymer types and application- Advanced materials (Biomaterials, nanomaterials and composites) with special reference to the applications in chemical Industries. |

**Text / Reference**

| T/R | Book Title/Author |
|---|---|
| T1 | Materials Science and Engineering an Introduction; William D Callister, David G Rethwisch |
| R1 | Materials Science and Engineering- A First course; V. Raghavan |

**Online Resources**

| Sl.No | Website Link |
|---|---|
| 1 | https://nptel.ac.in/chemical engineering |
| 2 | https://www.classcentral.com/course/edx-materials-science-and-engineering-8150 |
| 3 | https://www.edx.org/learn/materials-science |
| 4 | https://ocw.mit.edu/courses/materials-science-and-engineering/ |